Experiment - 6.1.2. Factorial of a Number

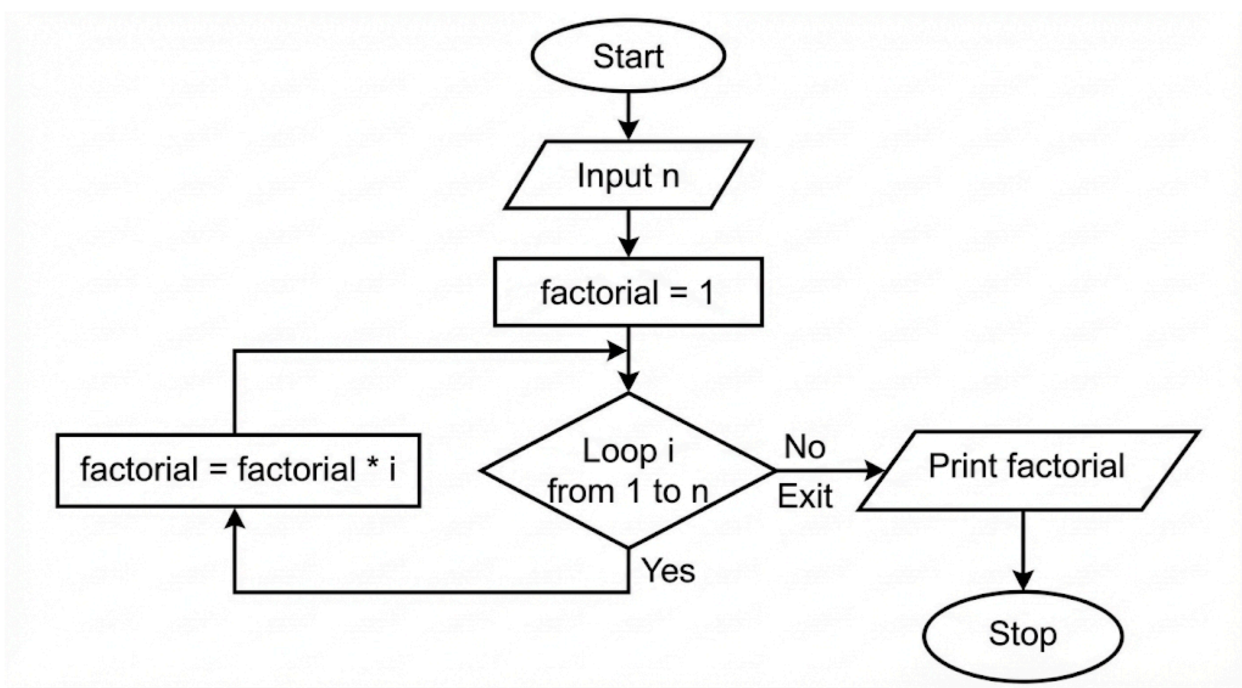
1. Aim

To design and implement a Python program that calculates the factorial of a given integer *n* using a for loop. The program reads the integer, computes its factorial iteratively, and prints the final result.

2. Pseudocode

1. **START**
2. **READ** a single integer from the user and **STORE** it in the variable n.
3. **INITIALIZE** a variable factorial to 1.
4. **FOR** each number i in the range from 1 to n (inclusive):
   - **MULTIPLY** factorial by i (factorial = factorial * i).
   - **STORE** the new value back into factorial.
5. **END FOR** loop once all numbers up to n have been multiplied.
6. **PRINT** the final value of factorial.
7. **END**

3. Flowchart



4. Python Program

```
n = int(input())

factorial = 1

for i in range(1, n + 1):
    factorial *= i

print(factorial)
```

5. Experiment Screenshot